# NETFLIX DATA SCIENCE & MACHINE LEARNING PROJECT

**Auspify Technologies | Data Science Internship**

## FINAL PROJECT REPORT

Six tasks: Data Cleaning, Exploratory Data Analysis, Recommendation System, Trend Prediction, Machine Learning Classification, and Business Insights Dashboard.

Prepared using the submitted Google Colab notebook, saved project outputs, and screenshots.

Dataset: Netflix catalogue metadata | 8,790 records | 10 original columns

Report scope: methodology, observed outputs, evaluation, limitations, and business recommendations.

# 1. Executive Summary

This internship project implements an end-to-end analysis of Netflix catalogue metadata. The workflow cleans data, explores patterns, recommends similar titles using metadata similarity, models historical title-count trends, classifies titles by content type, and presents results in an interactive Plotly dashboard. Findings are descriptive and experimental; the dataset contains catalogue metadata, not viewership or revenue.

# 2. Dataset and Technical Environment

The uploaded notebook uses a dataset of 8,790 rows and 10 original fields: show_id, type, title, director, country, date_added, release_year, rating, duration, and listed_in. The project uses Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Plotly, and Google Colab.

Content composition: 6,126 Movies (69.7%) and 2,664 TV Shows (30.3%).

# 3. Task 1: Data Cleaning and Preprocessing

The notebook checks nulls, placeholders, duplicate rows and identifiers, and parses date fields. A cleaned CSV was saved. Final notebook validation reports 8,790 records, zero duplicate rows, zero duplicate IDs, and zero invalid or missing dates. Some metadata remains missing: 1 title, 2,588 directors, and 287 countries. This should be disclosed rather than described as a fully complete dataset.



Figure 1. Final data quality check from the notebook.

# 4. Task 2: Exploratory Data Analysis

Visualizations examine the Movie/TV Show split, title release-year distribution, leading countries, and common genres. Country and genre fields may list multiple values, so aggregation depends on splitting or choosing the first listed category. The charts describe catalogue composition, not viewer preference.

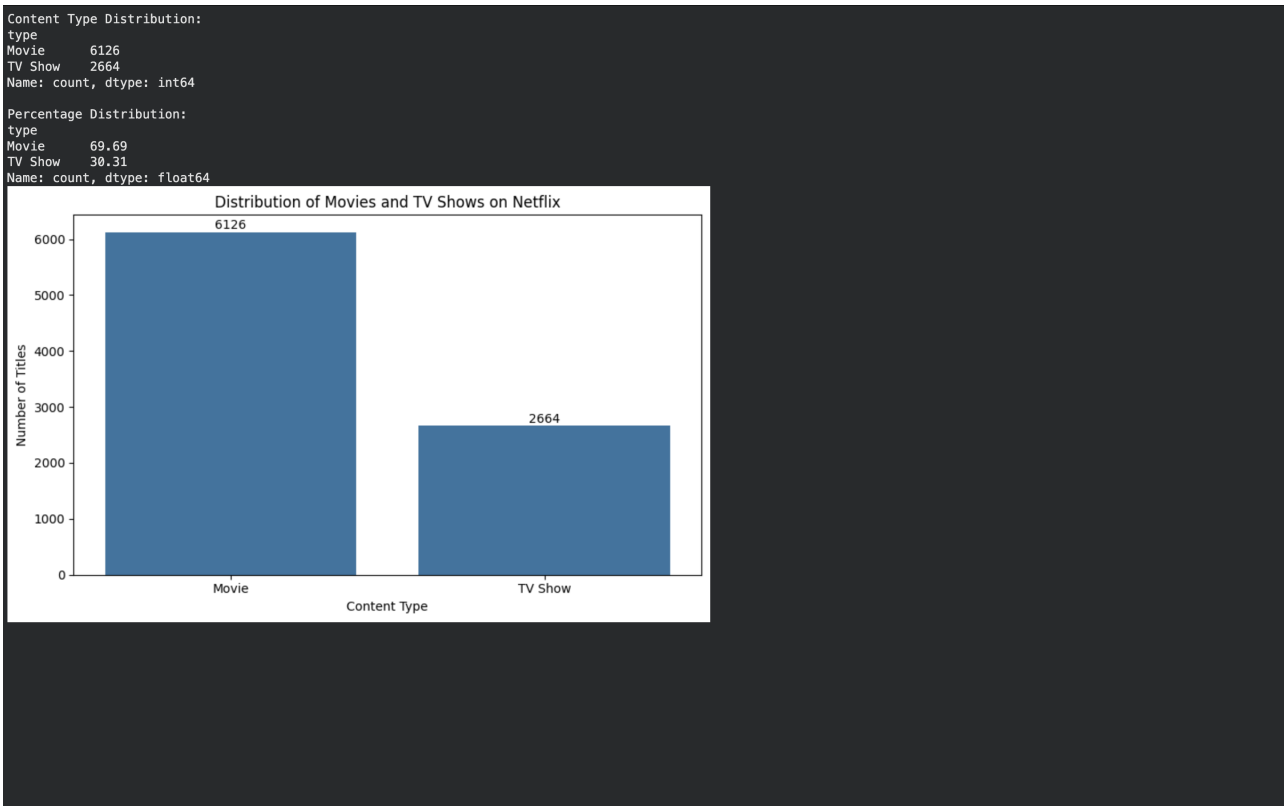
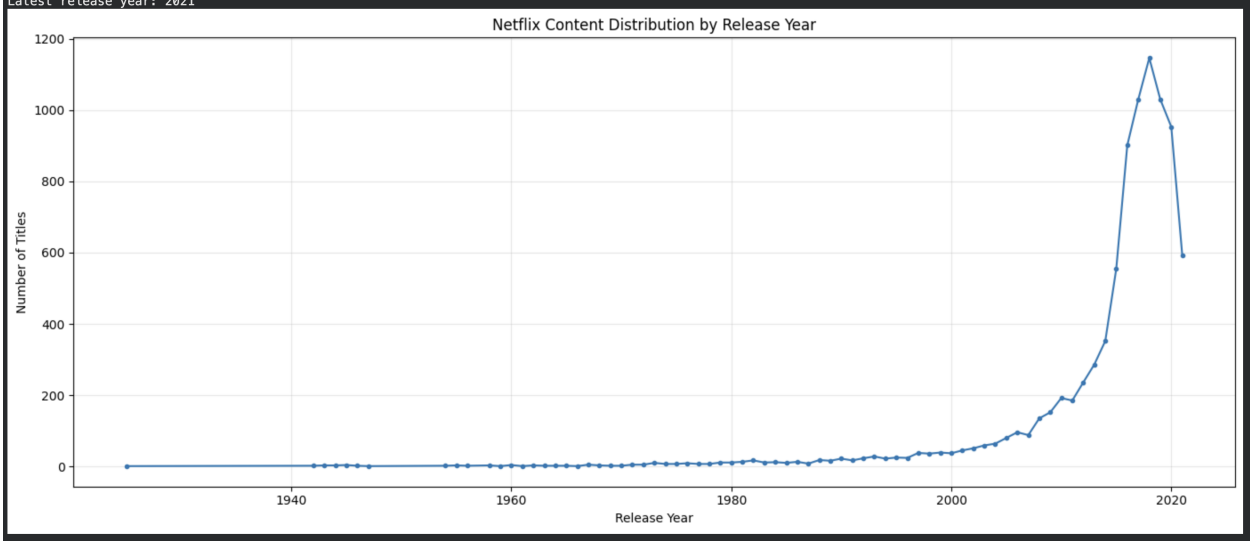



Figure 2. Movie versus TV Show distribution.

```
Earliest release year: 1925
Latest release year: 2021
```




```
Top 10 Release Years:
release_year
2018    1146
2017    1030
2019    1030
2020     953
2016     901
2021     592
2015     555
2014     352
2013     286
2012     236
Name: count, dtype: int64
```

Figure 3. Release-year trend.

```
Top 10 Genres/Categories:
listed_in
International Movies        2752
Dramas                      2426
Comedies                    1674
International TV Shows      1349
Documentaries                869
Action & Adventure           859
TV Dramas                    762
Independent Movies           756
Children & Family Movies     641
Romantic Movies              616
Name: count, dtype: int64
```

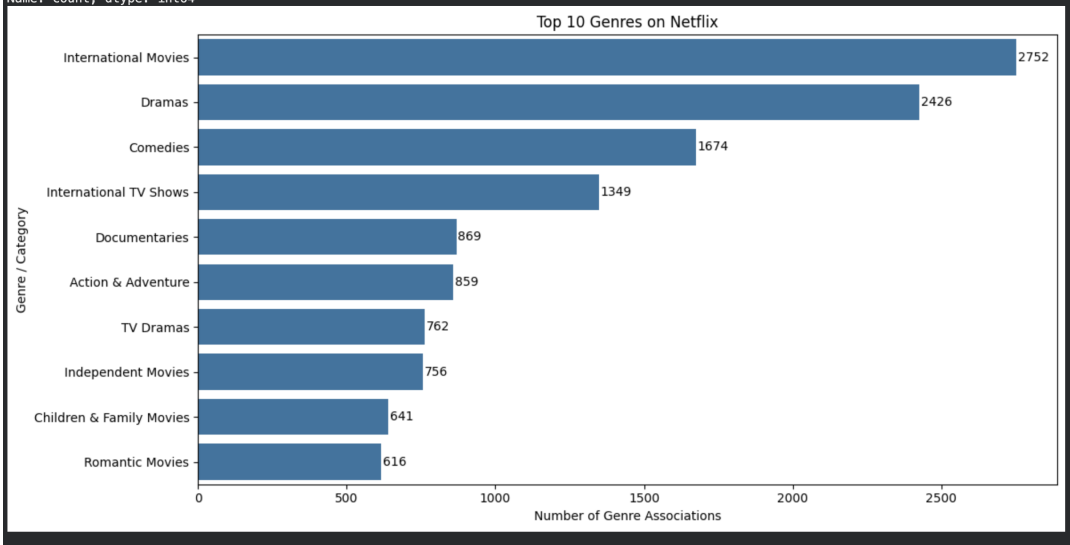
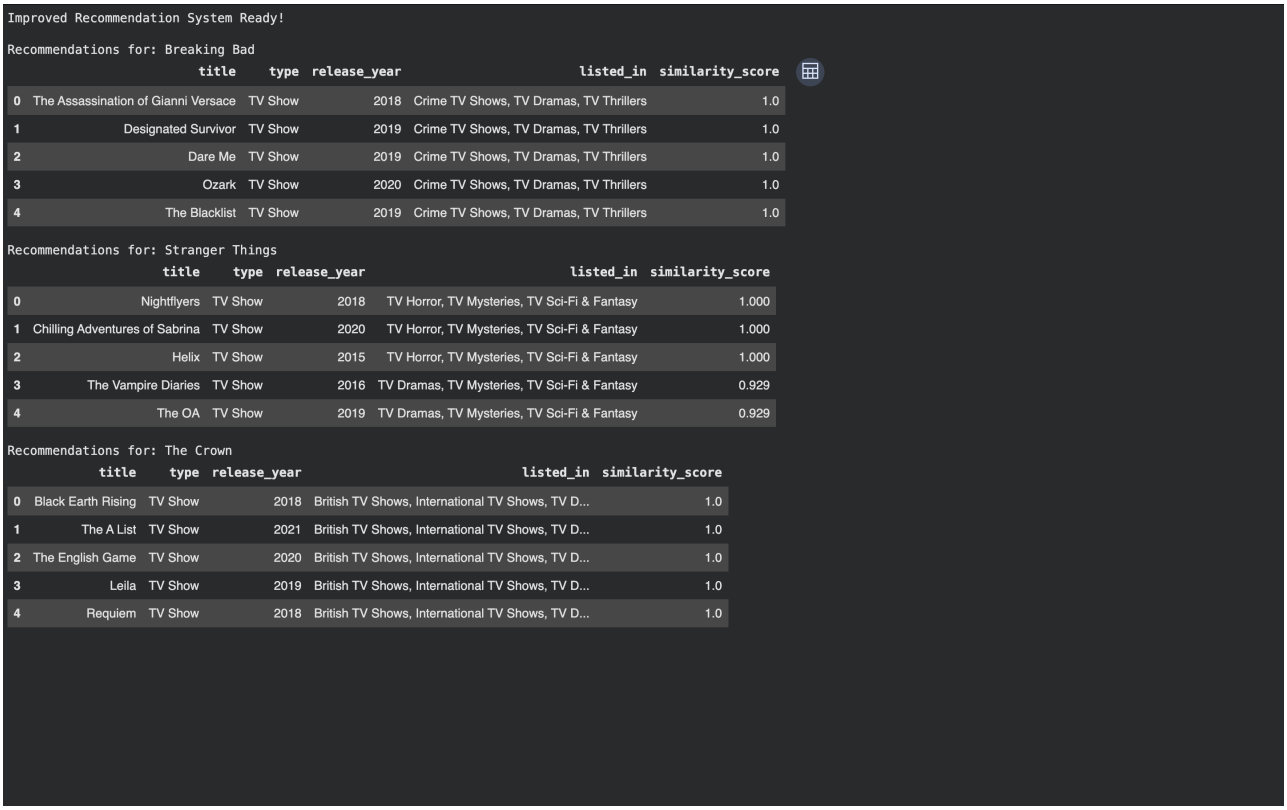



Figure 4. Top genres in the catalogue.

# 5. Task 3: Metadata-Based Recommendation System

The notebook builds TF-IDF feature representations from available fields such as type, director, country, rating, and genre, then uses cosine similarity to identify related titles. An improved version weights genre information more heavily. The first model reports 8,789 available titles and a TF-IDF matrix of 8,789 by 6,566. Example recommendations are generated for Breaking Bad, Stranger Things, and The Crown, with genre-overlap checks. Scores of 1.0 may reflect identical genre tags, not identical viewing appeal.



```
Improved Recommendation System Ready!

Recommendations for: Breaking Bad
```

| | title | type | release_year | listed_in | similarity_score |
|---|---|---|---|---|---|
| 0 | The Assassination of Gianni Versace | TV Show | 2018 | Crime TV Shows, TV Dramas, TV Thrillers | 1.0 |
| 1 | Designated Survivor | TV Show | 2019 | Crime TV Shows, TV Dramas, TV Thrillers | 1.0 |
| 2 | Dare Me | TV Show | 2019 | Crime TV Shows, TV Dramas, TV Thrillers | 1.0 |
| 3 | Ozark | TV Show | 2020 | Crime TV Shows, TV Dramas, TV Thrillers | 1.0 |
| 4 | The Blacklist | TV Show | 2019 | Crime TV Shows, TV Dramas, TV Thrillers | 1.0 |

```
Recommendations for: Stranger Things
```

| | title | type | release_year | listed_in | similarity_score |
|---|---|---|---|---|---|
| 0 | Nightflyers | TV Show | 2018 | TV Horror, TV Mysteries, TV Sci-Fi & Fantasy | 1.000 |
| 1 | Chilling Adventures of Sabrina | TV Show | 2020 | TV Horror, TV Mysteries, TV Sci-Fi & Fantasy | 1.000 |
| 2 | Helix | TV Show | 2015 | TV Horror, TV Mysteries, TV Sci-Fi & Fantasy | 1.000 |
| 3 | The Vampire Diaries | TV Show | 2016 | TV Dramas, TV Mysteries, TV Sci-Fi & Fantasy | 0.929 |
| 4 | The OA | TV Show | 2019 | TV Dramas, TV Mysteries, TV Sci-Fi & Fantasy | 0.929 |

```
Recommendations for: The Crown
```

| | title | type | release_year | listed_in | similarity_score |
|---|---|---|---|---|---|
| 0 | Black Earth Rising | TV Show | 2018 | British TV Shows, International TV Shows, TV D... | 1.0 |
| 1 | The A List | TV Show | 2021 | British TV Shows, International TV Shows, TV D... | 1.0 |
| 2 | The English Game | TV Show | 2020 | British TV Shows, International TV Shows, TV D... | 1.0 |
| 3 | Leila | TV Show | 2019 | British TV Shows, International TV Shows, TV D... | 1.0 |
| 4 | Requiem | TV Show | 2018 | British TV Shows, International TV Shows, TV D... | 1.0 |

Figure 5. Example recommendation output.

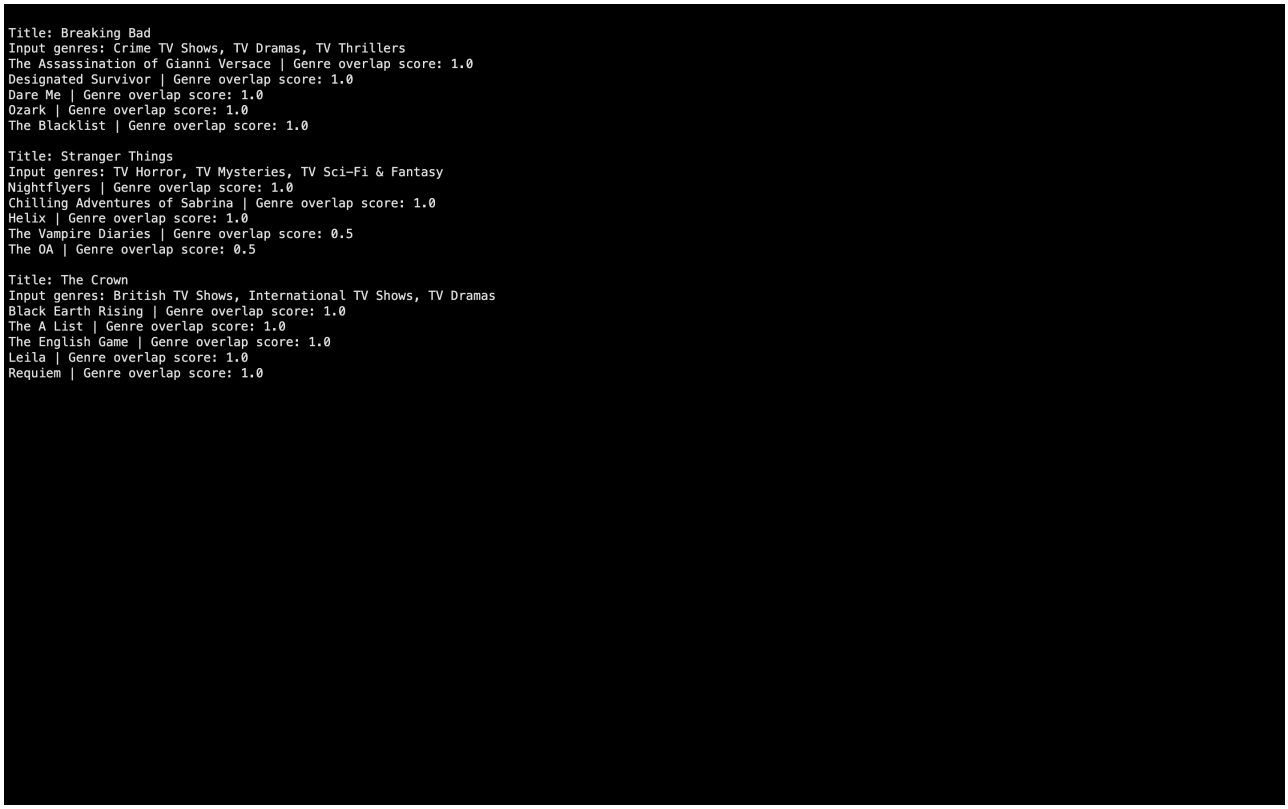


Figure 6. Genre overlap evaluation.

# 6. Task 4: Trend Prediction Analysis

Yearly counts based on release_year were modeled using Linear Regression and Random Forest Regression. Training used 2010–2018 and testing used 2019–2021. Linear Regression produced MAE 479.22 and RMSE 562.59. In the notebook model comparison, Random Forest achieved lower MAE (237.82) than Linear Regression (479.22). Forecasts are experimental extrapolations from a small historical sample; 2021 and later catalogue coverage may be incomplete, so future figures should not be interpreted as business commitments.

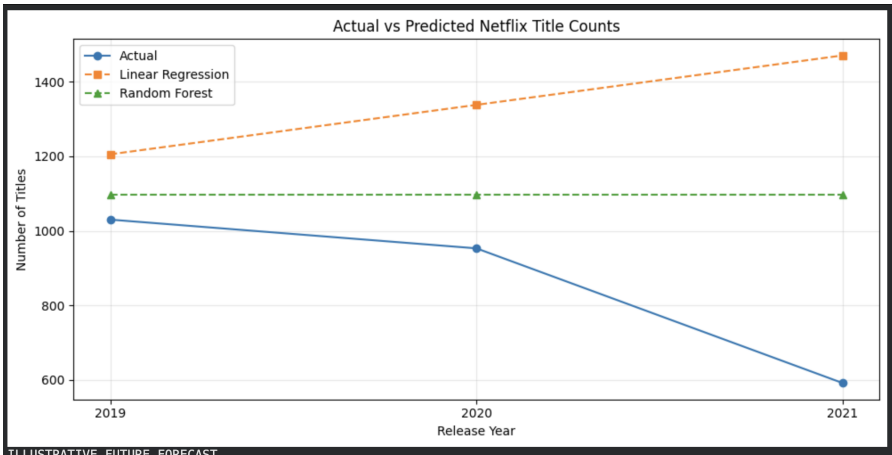
MODEL COMPARISON

| | Model | MAE |
|---|---|---|
| 0 | Linear Regression | 479.22 |
| 1 | Random Forest | 237.82 |

PREDICTION COMPARISON

| | Year | Actual | Linear Regression | Random Forest |
|---|---|---|---|---|
| 0 | 2019 | 1030 | 1205 | 1096 |
| 1 | 2020 | 953 | 1338 | 1096 |
| 2 | 2021 | 592 | 1470 | 1096 |

Figure 7. Forecast model comparison.




ILLUSTRATIVE FUTURE FORECAST

| | Year | Linear Regression | Random Forest |
|---|---|---|---|
| 0 | 2022 | 1603 | 1096 |
| 1 | 2023 | 1735 | 1096 |
| 2 | 2024 | 1868 | 1096 |
| 3 | 2025 | 2000 | 1096 |

Forecast CSV saved successfully!

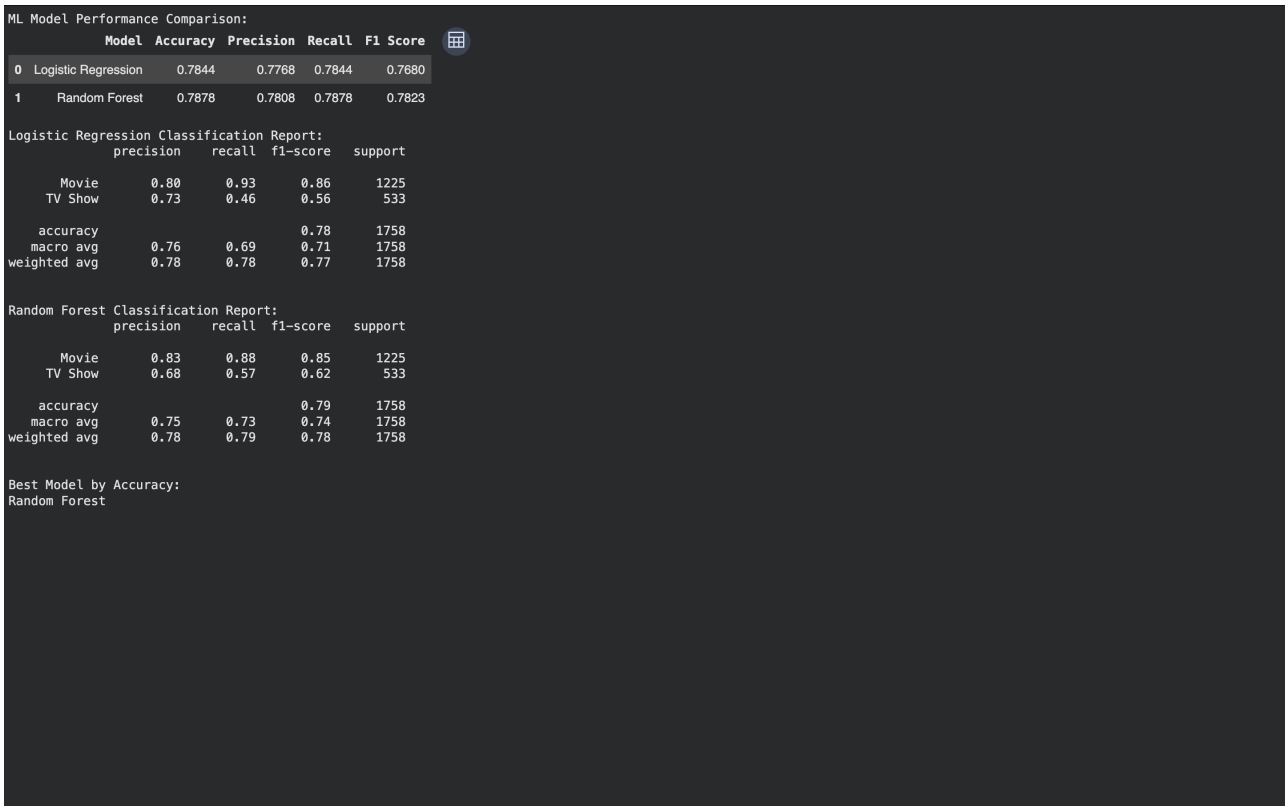
Figure 8. Historical fit and illustrative future forecast.

# 7. Task 5: Machine Learning Classification

The classification task predicts whether a title is a Movie or TV Show from release_year, country, rating, and year_added. The notebook uses a stratified 80/20 split: 7,032 training records and 1,758 test records. Numeric and categorical fields are preprocessed before fitting Logistic Regression and Random Forest classifiers.

| Model | Accuracy | Weighted precision | Weighted F1 |
|---|---|---|---|
| Logistic Regression | 78.44% | 77.68% | 76.80% |
| Random Forest | 78.78% | 78.08% | 78.23% |

Random Forest performs slightly better on these test metrics. Classification is based on catalogue metadata; these results should not be confused with a recommendation-system accuracy score.



ML Model Performance Comparison:

| | Model | Accuracy | Precision | Recall | F1 Score |
|---|---|---|---|---|---|
| 0 | Logistic Regression | 0.7844 | 0.7768 | 0.7844 | 0.7680 |
| 1 | Random Forest | 0.7878 | 0.7808 | 0.7878 | 0.7823 |

Logistic Regression Classification Report:

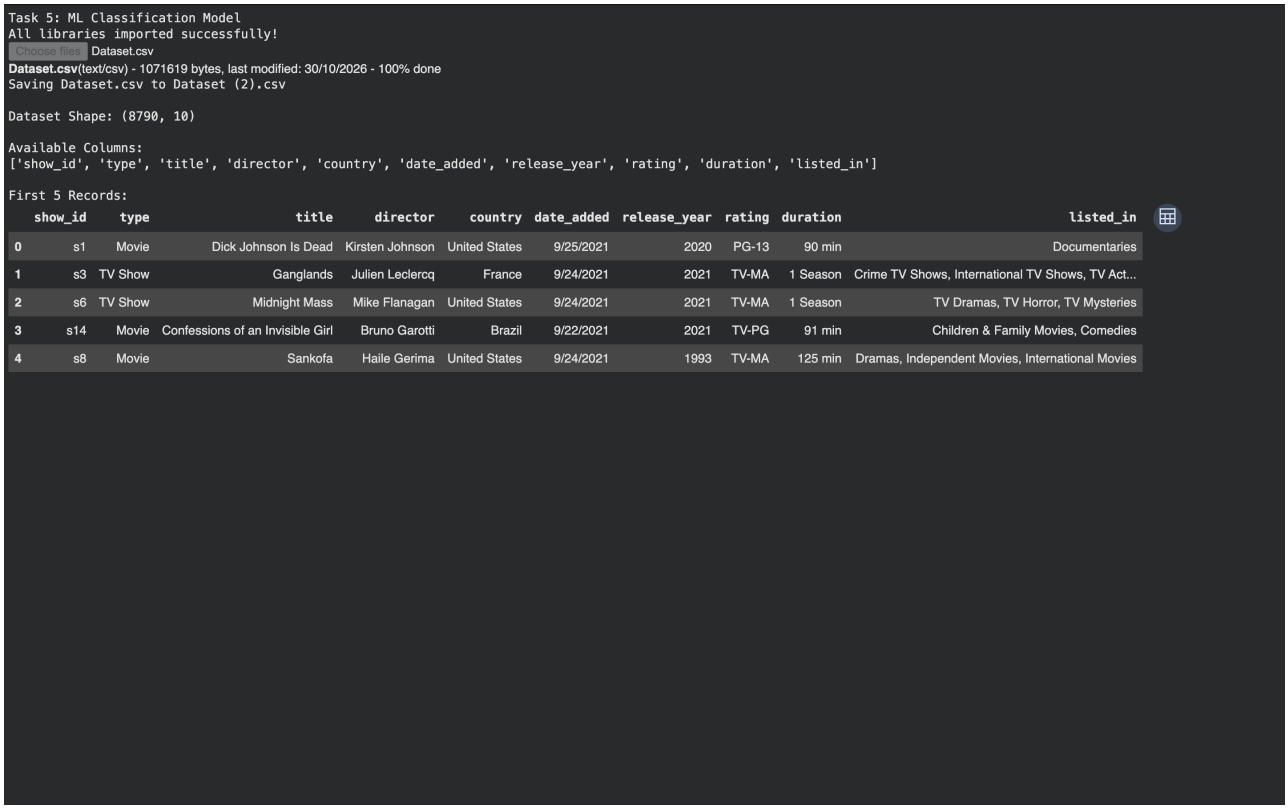
| | precision | recall | f1-score | support |
|---|---|---|---|---|
| Movie | 0.80 | 0.93 | 0.86 | 1225 |
| TV Show | 0.73 | 0.46 | 0.56 | 533 |
| accuracy | | | 0.78 | 1758 |
| macro avg | 0.76 | 0.69 | 0.71 | 1758 |
| weighted avg | 0.78 | 0.78 | 0.77 | 1758 |

Random Forest Classification Report:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| Movie | 0.83 | 0.88 | 0.85 | 1225 |
| TV Show | 0.68 | 0.57 | 0.62 | 533 |
| accuracy | | | 0.79 | 1758 |
| macro avg | 0.75 | 0.73 | 0.74 | 1758 |
| weighted avg | 0.78 | 0.79 | 0.78 | 1758 |

Best Model by Accuracy:
Random Forest

Figure 9. Classification performance results.




| | show_id | type | title | director | country | date_added | release_year | rating | duration | listed_in |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | s1 | Movie | Dick Johnson Is Dead | Kirsten Johnson | United States | 9/25/2021 | 2020 | PG-13 | 90 min | Documentaries |
| 1 | s3 | TV Show | Ganglands | Julien Leclercq | France | 9/24/2021 | 2021 | TV-MA | 1 Season | Crime TV Shows, International TV Shows, TV Act... |
| 2 | s6 | TV Show | Midnight Mass | Mike Flanagan | United States | 9/24/2021 | 2021 | TV-MA | 1 Season | TV Dramas, TV Horror, TV Mysteries |
| 3 | s14 | Movie | Confessions of an Invisible Girl | Bruno Garotti | Brazil | 9/22/2021 | 2021 | TV-PG | 91 min | Children & Family Movies, Comedies |
| 4 | s8 | Movie | Sankofa | Haile Gerima | United States | 9/24/2021 | 1993 | TV-MA | 125 min | Dramas, Independent Movies, International Movies |

Figure 10. Classification model visual outputs.

# 8. Task 6: Interactive Business Insights Dashboard

A standalone HTML dashboard combines five summary KPI cards with six interactive Plotly charts: content type, yearly additions, top countries, top genres, ratings, and ML model performance. The submitted notebook subsequently filters unknown country labels from the country chart. The dashboard loads Plotly via CDN and therefore needs internet access for full interactive rendering.

```
NETFLIX BUSINESS DASHBOARD — DATA READY
Total Titles: 8790
Total Movies: 6126
Total TV Shows: 2664
Countries: 86

Dashboard Dataset Preview:
```

| | show_id | type | title | director | country | date_added | release_year | rating | duration | listed_in | year_added | primary_genre | primary_country |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | s1 | Movie | Dick Johnson Is Dead | Kirsten Johnson | United States | 2021-09-25 | 2020 | PG-13 | 90 min | Documentaries | 2021 | Documentaries | United States |
| 1 | s3 | TV Show | Ganglands | Julien Leclercq | France | 2021-09-24 | 2021 | TV-MA | 1 Season | Crime TV Shows, International TV Shows, TV Act... | 2021 | Crime TV Shows | France |
| 2 | s6 | TV Show | Midnight Mass | Mike Flanagan | United States | 2021-09-24 | 2021 | TV-MA | 1 Season | TV Dramas, TV Horror, TV Mysteries | 2021 | TV Dramas | United States |
| 3 | s14 | Movie | Confessions of an Invisible Girl | Bruno Garotti | Brazil | 2021-09-22 | 2021 | TV-PG | 91 min | Children & Family Movies, Comedies | 2021 | Children & Family Movies | Brazil |
| 4 | s8 | Movie | Sankofa | Haile Gerima | United States | 2021-09-24 | 1993 | TV-MA | 125 min | Dramas, Independent Movies, International Movies | 2021 | Dramas | United States |

Figure 11. Dashboard overview.

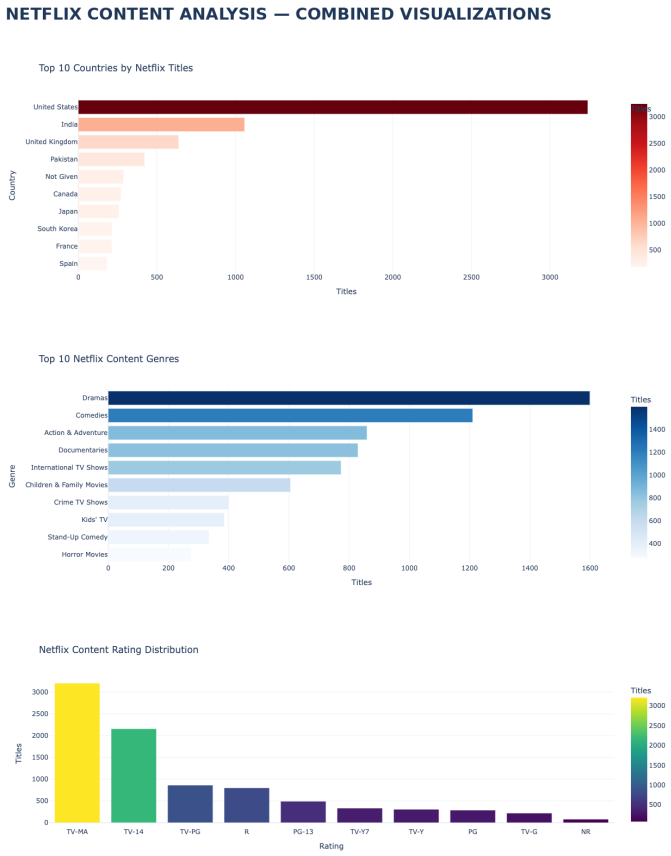



Figure 12. Country, genre, and rating analysis.



Figure 13. ML performance dashboard.

# 9. Business Insights and Recommendations

Catalogue composition is Movie-heavy, so portfolio planning can compare the mix of film and episodic content. Genre and country concentration can guide questions for content acquisition teams, but the dataset alone cannot establish audience demand. Historical additions can be monitored over time, with care for partial-year coverage. The classification experiment demonstrates predictive modeling and may support metadata-quality workflows after additional validation.

Recommended next steps: connect viewer engagement and regional demand data; standardize multi-valued country and genre fields; monitor data completeness; evaluate recommendation quality with human feedback or interaction data; and benchmark models on a genuinely held-out future sample.

# 10. Limitations and Ethics

The source contains no user-level viewing histories, satisfaction scores, subscriptions, or revenue. The recommendation system is content-based rather than collaborative filtering. Missing director and country values remain after cleaning. The time-series experiment has a short evaluation horizon, and the dashboard describes historical catalogue data rather than live Netflix operations. Avoid treating inferred geographic or genre popularity as measured audience demand.

# 11. Conclusion

The submitted notebook demonstrates six connected data science tasks, from data validation through predictive models and an interactive dashboard. Random Forest outperformed the alternative models on the reported trend MAE and classification accuracy comparisons. The strongest project outcome is the integrated, reproducible workflow and clear visual reporting; deployment or business adoption would require additional data and validation.

# 12. Submitted Evidence

Evidence reviewed: the uploaded .ipynb notebook, screenshots for Tasks 1–6, netflix_cleaned.csv, yearly trend and future forecast CSVs, and the Task 5 and Task 6 ZIP packages. The master ZIP did not include the original Dataset.csv or a README.md as separate top-level files; add them before final GitHub submission if available. Task 1 cleaning does not eliminate all missing metadata.